

AMIS1012 Ethics in Computing

Chapter 1

INTRODUCTION TO ETHICS

Course Outline

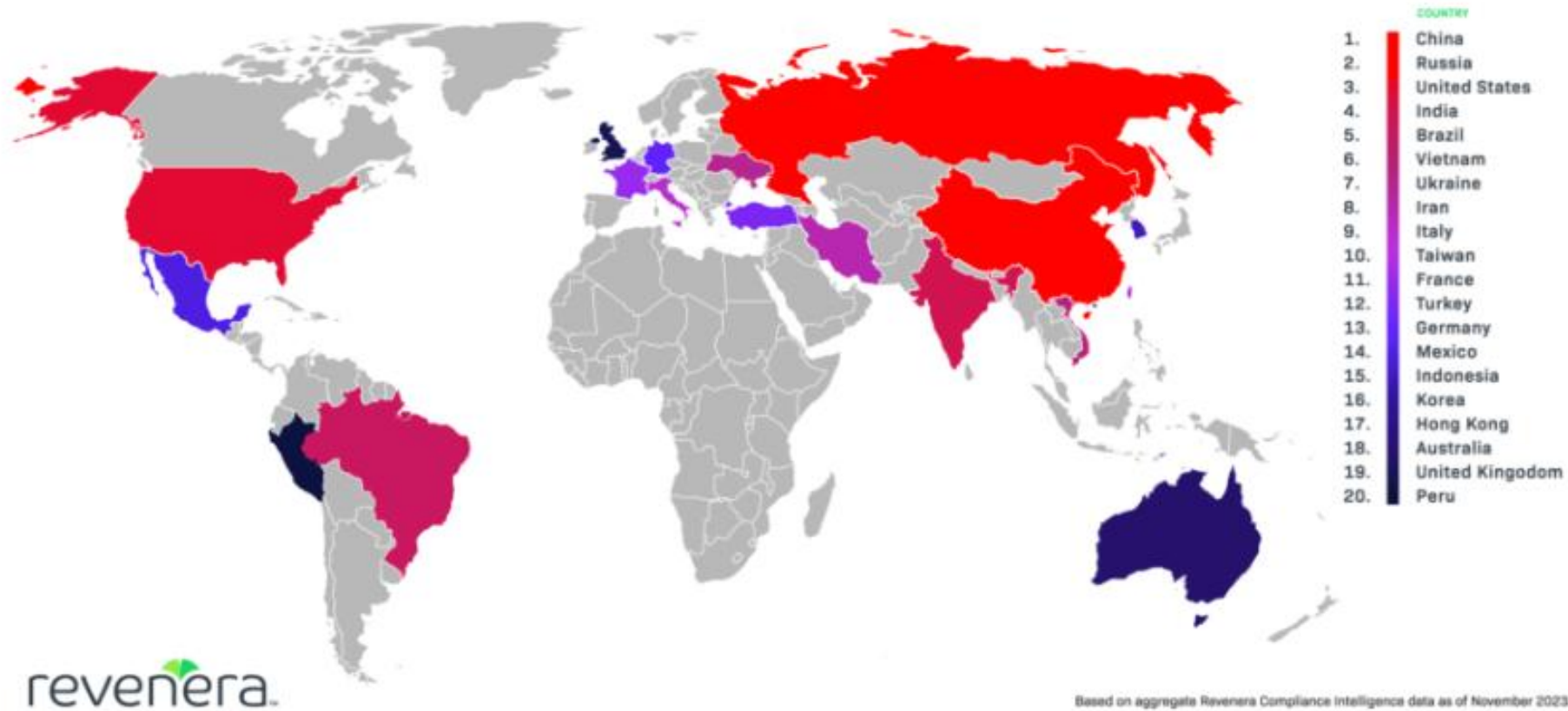
Introduction to Ethics

- Definition of Ethics.
- Ethical Consideration in Decision Making.
- Difference between Moral, Ethics and Laws.
- Requirement of Ethics in Information Technology.
- Ethics for IT Professionals.
- Common Ethical Issues for IT Users.
- Certifications.

Definition of Ethics

- **Ethics** is a set of values that define what is and is not acceptable behaviour in a certain society.
- **Computer ethics** are guidelines for the morally acceptable use of computers in the society.
- Ethical behaviour is described by a set of widely agreed standards, many of which are almost universal. Lies and stealing, for example, are immoral.
- Opinions of what constitutes ethical action differ dramatically. For example, software piracy, which is the act of making unauthorized copies of software, is considered ethical in some countries but not ethical in others.

Top 20 Software License Misuse and Piracy Hotspots



Definition of Ethics

- People often build patterns that help them distinguish between what culture finds good and poor.
- A virtue is a habit that encourages people to do what is right, and a sin is a habit that encourages people to do what is wrong.
- Virtues such as fairness and honesty are some of the examples of virtues, while envy and rage are examples of vices.

Definition of Ethics

- In other words, ethics is a subset of philosophy concerned with principles related to human behavior, specifically the rightness and wrongness of such acts, as well as the goodness and badness of the intentions and purposes of certain actions.

Ethical Consideration in Decision Making

- Steps involved in ethical decision making.
- Further reading: 1.3.1 until 1.3.5
Awari, G.K., & Warjekar, S.V. (2022).
Ethics in Information Technology: A
Practical Guide . CRC Press, Taylor &
Francis Group.
<https://tarc.idm.oclc.org/login?url=https://www.taylorfrancis.com/books/9781003280989>



Ethical Consideration in Decision Making

- Steps involved in ethical decision making (cont):

Steps	Description
Develop Problem Statement	A problem statement is a succinct, straightforward explanation of the problem that must be solved.
Identity Alternatives	It is optimal to engage the support of others, including stakeholders, at this point of decision-making to find multiple alternative solutions to the issue.
Evaluate & Choose Alternative	If a range of alternatives has been developed, the group aim to test them using a variety of metrics, including efficacy in solving the problem, the level of risk associated with each solution, cost, and implementation time.
Implement Decision	If a solution has been chosen, it must be executed in a timely, reliable, and effective manner.
Evaluation Results	Track the outcome to see whether the desired consequence was accomplished, and observe the effects on the company and different stakeholders since the problem solution has been applied.

Ethical Consideration in Decision Making

- Most ethical choices involve:
 - a. several options
 - b. long-term consequences
 - c. unpredictable outcomes
 - d. outcomes that combine different fiscal, legal, and social advantages and costs
 - e. personal ramifications
- Organizational ethics decisions are not black- and-white choices; they require deeper considerations weighing economic and self-interest benefits and costs against different legal, ethical, and social benefits and costs.

Difference between Moral, Ethics and Laws

- **Morals** refer to one's religious views on what is right and wrong.
- **Ethics** refers to the rules or codes of conduct that a society (nation, institution, or profession) expects of an individual.
- **Law** is a set of laws that determines what we are permitted to do and what we are not permitted to do.

Difference between Moral, Ethics and Laws

- A moral action backs up what an individual believes is the right thing to do. Laws will declare an act lawful, even though certain people believe it is immoral (Example: abortion)
- For example, the principles of the legal profession require lawyers to protect an innocent defendant to the best of their abilities, particularly though they are well aware that the client has committed the most terrible and morally repugnant crime conceivable.

Difference between Moral, Ethics and Laws

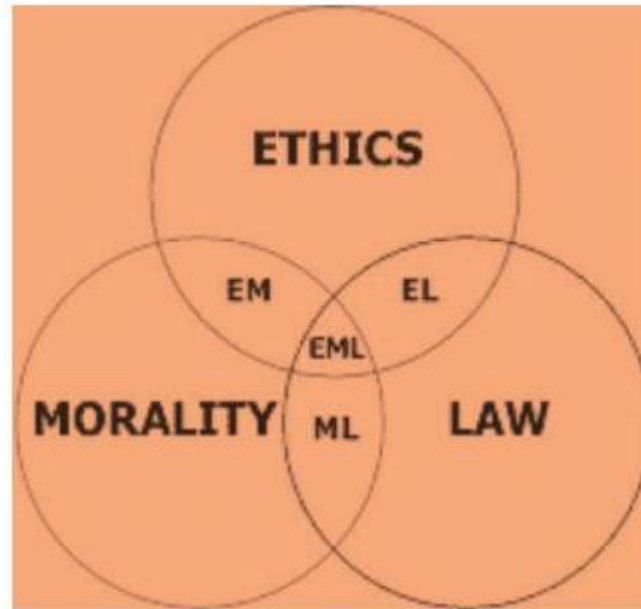


Figure 1: Differences between morals, ethics, and laws.

Requirement of Ethics in Information Technology

- The implementation of information systems is intertwined with social and political affairs, making ethical considerations of how information is handled even more important.
- Electronic devices have infiltrated all aspects of governance, work, and personal life to the point that even those without access to them have had a huge impact on how they operate.

Requirement of Ethics in Information Technology

- **Electronic knowledge system ethical issues**
 - The regulation of electronic records and access to information, privacy and data manipulation, and international implications are all broad concerns.
 - All of these apply to computer networks, electronic databases, and geographic information systems in particular.

Requirement of Ethics in Information Technology

- **Privacy policy for computer ethics**
 - Less obvious types of machine activity may pose ethical concerns.
 - **For example:** Employees' use of computers for personal purposes. Most workers who use laptops have an email address and review it often at work. In general, the company's important internal messages are often delivered by email. Employees, on the other hand, will be given a personal email address and may use the business computer to send and receive personal messages

Requirement of Ethics in Information Technology

- **Privacy policy for computer ethics**
 - New technologies permit not only the checking of email, but even other Internet practices such as message boards, text, and even web browsing.
 - While businesses will want to ensure that their workers spend their time with the company, track network traffic, and catch ethical lapses, this is difficult to do because the motivations for using a website cannot be decided merely by recognizing that someone visited it.

Requirement of Ethics in Information Technology

- **Importance of ethics in the field of information and technology**

With the proliferation of the Internet, the opportunity to collect and retain large volumes of personal information, and high dependence on computer systems, the possibility of information technology being unethically has risen.

Requirement of Ethics in Information Technology

Scenarios that raise public concern:

Scenarios	Explanation
Tracking employee email and Internet usage	The email and Internet use of many employees is tracked when they are at work, while managers want to strike a balance between their ability to handle vital self-direction and their need to control productivity.
Downloading of free music and movies	Millions of people have downloaded free music and movies, ostensibly in breach of copyright rules, at great cost to the copyright owners.
Spamming	Organizations use unsolicited email (spam) to reach millions of users around the world as a low-cost communications strategy

Requirement of Ethics in Information Technology

Scenarios that raise public concern (cont.):

Scenarios	Explanation
Hackers and identity fraud	Hackers gain access to the files of financial and retail institutions in order to harvest consumer records, which they then use to conduct identity fraud by creating new identities and making payments on the credit cards of unsuspecting suspects.
Plagiarism	Students from all over the world have been found plagiarizing content for their term papers by copying material from the Internet.
Cookies and spyware	To log users online transactions and events, websites place cookies or spyware on their hard drives

Ethics for IT Professionals

- The information technology industry is said to be the world's fastest developing and most diverse industry. Computers have become an integral part of everyone's lives, and they are used in a variety of fields including education, networking, industry, entertainment, building, medicine, and security.
- Information technology is used to create communications campaigns for entrepreneurs, accounting software for financial firms, database creation for effective networking, resource control, and customer service, as well as the development of technologies and equipment for agriculture, security, medicine, engineering, and other fields. Simply said, IT Professionals are those who work in the field of information technology.

Ethics for IT Professionals

Characteristics of a good tech career:

Characteristics	Explanation
Patience	IT professional resources necessitate continuous analysis, trial, and error attempts to arrive at a specific solution. As a result, one must be patient with the individuals and equipment with which one operates. Before coming to any conclusions, the dilemma should be carefully examined, and hasty assumptions and assessments should be avoided.
Able to communicate effectively	When engaging with various types of clients, he/ she should be understandable, clear, and concise. It is important to have expertise and experience dealing with various styles of individuals using various communication techniques.
Able to adapt quickly	The area of IT technical definition is constantly changing and evolving. One must be able to adapt quickly and have an ability to explore new technology and applications.

Ethics for IT Professionals

Characteristics of a good tech career:

Characteristics	Explanation
Multitasking	IT professional should be able to do several tasks at the same time.
Problem-solving ability	Once encounter an unfamiliar problem, IT professionals need to use his/ her own problem-solving philosophies and methods, as well as the scientific skills he/ she has gained from work experience and study.
Excited about the job	An IT professional should be excited about the job and willing to learn more about the industry and be able to apply what he or she has learned.

Ethics for IT Professionals

- **Professionals**

An individual “working in a professional career capacity” is described by the US Code of Federal Regulations as someone who meets the following four criteria:

1. Doing work that necessitates advanced expertise in an area of science or engineering typically gained through a lengthy course of specialized scientific instruction and research or work.
2. The instruction, research, or practice is unique and imaginative in a recognized area of artistic achievement. The work outcome is solely dependent on the employee’s professional invention, imagination, or ability.

Ethics for IT Professionals

- **Professionals (cont.)**

3. Consistent application of discretion and judgment.
4. Mostly intellectual and diverse in nature and the performance career or outcome cannot be standardized over a specific time frame.

Ethics for IT Professionals

- **Professionals (cont.)**

- Many individuals, especially professionals, are expected to contribute to society and participate in a program of lifelong learning (both formal and informal), to stay current in their career and to help other professionals develop.

Ethics for IT Professionals

- **Professional code of ethics**

- A professional code of ethics outlines the ideals and standards that are fundamental to a certain occupational group's practice.
- Most occupations have a code of ethics that regulates how they conduct themselves. Most technical organization's codes of ethics are divided into two sections:
 - First: specifies the organization's goals.
 - Second: the guidelines and standards that all members are supposed to follow.

Ethics for IT Professionals

- Laws should not serve as a comprehensive roadmap to ethical conduct. But, adhering to a professional code of ethics will bring a number of advantages for individuals, professions, and society as a whole:

Ethical conduct	Advantages
Ethical decision-making	Adherence to a professional code of ethics requires that professionals make ethical decisions based on a shared collection of core principles and convictions.
High ethical principles and practice	Professionals who follow a code of ethics are reminded of the responsibilities and roles they may be tempted to breach in order to meet the demands of day-to-day business. To instruct practitioners in their relationships with others, the code also specifies permissible and inappropriate practices.
Public interest and confidence	Public trust is based on the presumption that a practitioner can act ethically. People often rely on a professional's honesty and sound judgment to say the truth, refrain from providing self-serving advice, and provide warnings about the possible negative consequences of their decisions. As a result, following a code of ethics increases faith and respect for practitioners and their field.
Self-evaluation benchmark	A code of ethics serves as a self-evaluation benchmark for professionals. The code of ethics may also be used by the professional's peers for praise or criticism.

Ethics for IT Professionals

- Seven trends are reshaping the professional services industry

Trends	Explanation
Client sophistication	Clients are more aware of what they need from service providers, are more inclined to go outside their own organization for the appropriate benefits, and are more capable of negotiating hard to receive the finest services at the lowest feasible cost.
Governance	Major scandals, as well as stricter legislation enforced to prevent further scandals (e.g., The Sarbanes-Oxley Act (SOX) is a U.S. federal law enacted in 2002 to enhance corporate governance, financial transparency, and the integrity of financial reporting in publicly traded companies), have created an atmosphere in which clients and service providers have less confidence and more oversight.
Interconnectedness	Clients and service providers have established their business ties on the expectation of a positive outcome that is being able to communicate easily and quickly anywhere in the world through electronic teleconferences, audio conferences, email, and wireless devices.

Ethics for IT Professionals

- Seven trends are reshaping the professional services industry (cont.)

Trends	Explanation
Discretion	Clients want to monitor and manage work-in-progress in real time. Clients are no longer able to wait until the final product is complete before providing input and comments.
Modularization	Clients should deconstruct their business processes into their basic steps and determine which ones they can handle directly and which ones they can outsource to service providers.
Globalization	Clients will compare and choose from a wide range of service providers all around the world, rendering the service provider market highly competitive.
Commoditization	Low-end service delivery (e.g., personnel augmentation to complete a project) is seen by clients as a commodity service when the main criteria for choosing a service provider is pricing. Clients aim to establish partnerships with their service providers in order to provide high-end offerings (e.g., the implementation of an IT business plan).

Common Ethical Issues for IT Users

- **Software Piracy**

- In a business environment, software piracy may often be traced back to IT experts, who can either encourage or actively participate in it. Corporate IT access practices and administration should enable consumers to report and contest cases of piracy.

Common Ethical Issues for IT Users

- **Inappropriate use of computing resources**
 - Many workers use their laptops to visit popular websites unrelated to their careers, engage in chat rooms, and play video games. These activities consume resources and reduce job productivity.
 - Furthermore, behaviors such as sharing inappropriate contents, jokes, and sending hate email could result in complaints and charges that an organization allowed a racist or sexual harassment-friendly work atmosphere.

Common Ethical Issues for IT Users

- **Inappropriate sharing of information**
 - Business has a huge volume of data that can be categorized as either private or confidential. A worker's private data include items such as wage information, attendance reports, health records, and performance reviews.
 - Private data also contain details for consumers, such as payment card numbers, phone numbers, and home addresses. Confidential information include sales and marketing calendars, manufacturing processes, formulation of formulas, tactical and strategic strategies, and research and development.

Common Ethical Issues for IT Users

- **Inappropriate sharing of information (cont.)**
 - Perhaps accidentally, an IT consumer who communicates this information with an unauthorized party has compromised someone's privacy or generated the possibility of business information falling into the hands of rivals.

Common Ethical Issues for IT Users

- When establishing an IT use policy, businesses may do one or more of the following things:

Guidelines	Explanation
Establishing guidelines for use of company software	IT administrators at businesses must provide specific guidelines for the usage of personal computers and applications. Any firms establish agreements with technology providers to include PCs and software such that IT staff can work from home. Other firms assist workers with acquiring lawful copies of all the software they need to be profitable, whether they operate in a workplace, on the road, or at home.
Defining and limiting the appropriate use of IT resources	Employees must be encouraged to respect company IT resources and utilize them to improve their work performance by developing, communicating, and enforcing written standards. Employees are allowed limited personal usage while being prohibited from accessing undesirable websites or sending rude or harassing messages through corporate email.
Structuring information systems to protect data and information	Organizations must establish procedures and standards to limit data access to just those employees who need it. For example: through a corporate network, sales managers may have complete access to sales and marketing datasets, but their access can be restricted to items over which they have responsibility. If they do not require access to things such as research and development results, they can be refused access.

Certifications

- Certificate is awarded by a certifying body to a practitioner has a certain range of talents, expertise or ability.
- Certifications are offered by a variety of corporations and educational associations. Different certifying bodies have different views on the merit.
- Many companies (employer) regard them as a barometer of mastery of a certain range of fundamental skills. Some hiring managers, on the other hand, are skeptical of certifications because they are no substitution for experience and do not guarantee that an individual can perform well on the job.

Certifications

- Most IT professionals want to improve their talents, and qualification is a formal means to do so.
- Completing a credential, offers strong appreciation for these individuals, as well as a plan to help them develop and succeed in their professions.
- Certain certifications boost a worker's salary and career prospects, which is essential for tightly identified roles within a larger position. It necessitates the passing of a written test, which usually entails certification

Certifications

- **Vendors certification**

- Professional certifications offered by vendors in the industry, usually to train IT professionals to configure, implement, manage and troubleshoot the vendors products or services.
- Example:
 - CISCO certifications <https://www.cisco.com/site/us/en/learn/training-certifications/certifications/index.html>
 - Microsoft certifications https://learn.microsoft.com/en-us/credentials/browse/?credential_types=certification
 - AWS certifications <https://aws.amazon.com/certification/>

Certifications

- **Industry association certification**
 - The importance of these certifications vary widely based on where individuals are in their careers, what other certifications they have, and the essence of the IT work market.
 - Individuals seeking certification must normally have the necessary qualifications and experience, sit for and pass an exam, and adhere to and follow a code of ethics defined by the institution offering the certification.
 - They must pay a yearly fee, complete continuing education credits, and, in certain instances, pass a monthly or yearly renewal test to maintain their license.

Certifications

- **Industry association certification (cont.)**

- Industry association certifications often require more expertise and a wider outlook than manufacturer certifications
- IT certification is shifting away from strictly technological content and toward a wider mix of academic, industry, and behavioral competencies that are expected in today's challenging IT roles. .
- This pattern can be seen in business organization certifications for diverse positions including project management and network security
- Example of Industry Association Certification:
 - International Information System Security Certification Consortium, or (ISC)² Certifications
<https://www.isc2.org/certifications>
 - ISACA (Information Systems Audit and Control Association) Certifications:
<https://www.isaca.org/credentialing/certifications>

Certifications

- Examples of entry-level IT certifications:
 - 16 best entry-level IT certifications to launch your career (2023), CIO.com, <https://www.cio.com/article/222343/10-best-entry-level-it-certifications-to-launch-your-career.html>

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